Instruction Set Architecture (ex. ISA)

Exercises Computer Architecture

1 Assembly Language

1.1 C-Code to RISC-V Assembler

Compile the following C-Code into RISC-V assembler.

a)
```
a = b + c;
```

b)
```
a = b + c - d;
```

c)
```
a = b + 6;
```

d)
```
// int is a 32-bit signed word
int a = -372;
int b = a + 6;
```

e)
```
int a = 0xFEDC8765;
```

f)
```
int a = 0xFEDC8EAB;
```

1.2 C-Code to RISC-V Assembler

Compile the following C-Code into RISC-V assembler.

a)
```
if (i == j){
  f = g + h;
}
f = f - i;
```

b)
```
if (i == j){
  f = g + h;
}
else {
  f = f - i;
}
```

c)
```
// add the numbers from 0 to 9
int sum = 0;
int i;

for (i=0; i!=10; i=i+1){
  sum = sum + i;
}
```

d)
```
// add the powers of 2 from 1 to 100
int sum = 0;
int i;

for (i=1;i<101; i=i*2){
  sum = sum + i;
}
```

e)
```
int array[5];
array[0] = array[0] * 2;
array[1] = array[1] * 2;
```

f)
```
int array[1000];
int i;

for (i=0; i<100; i=i+1){
  array[i] = array[i] * 8;
}
```

g)
```
char str[80] = "CAT";
int len = 0;

// compute length of string
while (str[len]) len++;
```

1.3 RISC-V Assembler Execution

Execute the Assember code and specify the contents of the destination register rd if the source registers rs contain the following data:

$$s1 = 0x46A1\,F1B7$$

$$s2 = 0xFFFF\,0000$$

a) `and s3, s1, s2`

b) `or s4, s1, s2`

c) `xor s5, s1, s2`

1.4 RISC-V Assembler Execution

Execute the Assember code and specify the contents of the destination register rd, if the source registers rs contain the following data:

$$t3 = 0x3A75\,0D6F$$

a) `and s5, t3, -1484`

b) `or s6, t3, -1484`

c) `xor s7, t3, -1484`

1.5 RISC-V Assembler Execution

Execute the Assember code and specify the contents of the destination register rd, if the source registers rs contain the following data:

$$s1 = 0x4000\,0000$$

$$s2 = 0x8000\,0000$$

```
mulh s4, s1, s2
mul  s3, s1, s2
```

1.6 RISC-V Assembler Execution

Execute the Assember code and specify the contents of the destination register rd, if the source registers rs contain the following data:

$$s1 = 0x0000\,0011$$

$$s2 = 0x0000\,0003$$

```
div s3, s1, s2
rem s4, s1, s2
```

2 Machine Language

2.1 Machine code to RISC-V Assembler

Decode the following Machine Code into RISC-V assembler.

a) 0x41FE 83B3

b) 0xFDA4 8393